

Hydrological & Infrastructure Report

Rishi Ganga & Dhauliganga River Disaster (February 2021 Data)

This report compiles hydro-meteorological, flood forecasting, and infrastructural data corresponding to the February 2021 Chamoli disaster. It is designed to serve as boundary conditions and structural parameters for 1D/2D hydrodynamic simulations (e.g., Delft3D, HEC-RAS, SPH) for your SIH project.

1. Hydrological Data & Flood Forecasting (Feb 2021)

February Baseline: Being the dry winter season in the Himalayas, the baseline river discharge and reservoir levels were significantly lower than monsoon levels. The catastrophic event on February 7, 2021, caused an extreme transient spike in the hydrograph.

River Discharge & Gauge Station Data

The Central Water Commission (CWC) maintains gauge & discharge (G&D) sites downstream on the Alaknanda river at Joshimath, Karanprayag, and Rudraprayag. Because the upstream projects lacked robust early-warning discharge gauges, the most accurate temporal discharge data comes from Joshimath.

Time Period (Feb 2021)	Location	Observed Discharge (m³/s)	Remarks / Flood Forecasting Data
Feb 1 - Feb 6	Rishi Ganga / Dhauliganga	3.0 - 5.5	Pre-event baseline. Normal dry season snowmelt and baseflow.
Feb 7 (~10:30 AM)	Rishi Ganga Confluence	11,000 - 15,000	Estimated peak debris-flow discharge. Fluid depth rose 20-25m.
Feb 7 (~11:30 AM)	Joshimath G&D (CWC)	3,200 - 4,000	Downstream attenuated peak. Water level jumped by 5.2 meters.
Feb 8 - Feb 28	Joshimath G&D (CWC)	15.0 - 35.0	

			Post-event flow stabilized, slightly elevated due to localized landslide dam seepage.
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2. Dam Details & Reservoir Levels

The disaster impacted two major run-of-the-river (RoR) hydroelectric projects (HEPs). Complete structural parameters are tabulated below for inclusion in morphological or breach model inputs.

Rishi Ganga Small Hydroelectric Project (Upstream)

Parameter	Details
Location	Raini Village, Chamoli, Uttarakhand (~2000m amsl)
Project Type & Capacity	Run-of-the-River (RoR) Diversion Weir / 13.2 MW
Dam / Weir Height	Approx. 15 meters
Storage Capacity	Minimal pondage (< 0.1 Million Cubic Meters)
Reservoir Level (Feb 7)	Operating at crest level (~1,970m)
Spillway / Structure	Overflow weir, completely destroyed in the event (100% breach).

Tapovan Vishnugad Hydroelectric Project (Downstream)

Operated by NTPC, this 520 MW project is located approx. 8 km downstream from the Rishiganga HEP on the Dhauliganga river.

Parameter	Details
Location	Dhauliganga River, Tapovan, Chamoli (30°29'38.9" N, 079°37'39.1" E)
Dam Type & Height	Concrete Gravity Barrage / 22 meters above riverbed

Barrage Length	200 meters overall structure length
Storage Capacity	Live Storage: 0.57 MCM Gross Pondage: Approx. 0.85 MCM
Reservoir Levels	Full Reservoir Level (FRL): 1,803.5 m Min Draw Down Level (MDDL): 1,790.5 m
Spillway / Gates	4 bays of 12m each. Gates measuring 12.3m high and 9.2m wide (or 14x12m overall openings).
Feb 2021 Status	Under construction, heavily silted and severely damaged.

3. Input Data Sourcing & References

For high-fidelity simulations, the data provided above should be correlated with:

- **India-WRIS / CWC Portals:** Hourly gauge data from the Joshimath site for Feb 2021.
- **National Register of Large Dams (NRLD):** Cross-referenced structural dimensions for the Tapovan Vishnugad barrage.
- **NDMA/CWC Special Report:** "Detailed Report-Study of Causes & Impacts of the Uttarakhand Disaster on 7th Feb 2021."